Olympiad Algebraic Manipulation, Vieta & Inequalities Practice Exercises

1 Let x be a positive real number satisfying $x + \frac{1}{x} = 3$ (illustrated on the positive branch of $g(x) = x + 1/x$). Without solving for x, find the exact value of $x^3 + \frac{1}{x^3}$.

a. 27

b. 36

c. 18

d. 15

2 Let x be a positive real number satisfying $x + \frac{1}{x} = 4$ (illustrated on the positive branch of $g(x) = x + 1/x$). Without solving for x, find the exact value of $x^4 + \frac{1}{x^4}$.

a. 196

b. 194

c. 198

d. 190

3 Let x be a positive real number satisfying $x + \frac{1}{x} = 5$ (illustrated on the positive branch of $g(x) = x + 1/x$). Without solving for x, find the exact value of $x^3 + \frac{1}{x^3}$.

a. 110

b. 125

c. 140

d. 105

4 Let x be a positive real number satisfying $x + \frac{1}{x} = 6$ (illustrated on the positive branch of $g(x) = x + 1/x$). Without solving for x, find the exact value of $x^4 + \frac{1}{x^4}$.

a. 1156

b. 1158

c. 1148

d. 1154

5 Let r, s, t be the three complex roots of the cubic polynomial $P(z) = z^3 - 3z^2 + 2z - 1$. Using Newton's power-sum recurrences, find $r^3 + s^3 + t^3$.

a. 9

b. 15

c. 12

d. 13

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6 Let r, s, t be the three complex roots of the cubic polynomial $P(z) = z^3 - 4z^2 + 3z - 2$. Using Newton's power-sum recurrences, find $r^3 + s^3 + t^3$.

a. 28

b. 34

c. 40

d. 35

7 Let r, s, t be the three complex roots of the cubic polynomial $P(z) = z^3 - 5z^2 + 4z - 3$. Using Newton's power-sum recurrences, find $r^3 + s^3 + t^3$.

a. 74

b. 65

c. 85

d. 75

8 Let r, s, t be the three complex roots of the cubic polynomial $P(z) = z^3 - 6z^2 + 5z - 4$. Using Newton's power-sum recurrences, find $r^3 + s^3 + t^3$.

a. 126

b. 156

c. 139

d. 138

9 Let x, y, z be positive real numbers satisfying $x + y + z = 1$. Find the exact minimum value of $\frac{1}{x} + \frac{4}{y} + \frac{9}{z}$.

a. 14

b. 42

c. 36

d. 34

10 Let x, y, z be positive real numbers satisfying $x + y + z = 7$. Find the exact minimum value of $\frac{1}{x} + \frac{4}{y} + \frac{16}{z}$.

a. 21

b. 7

c. 14

d. 5

11 Let x, y, z be positive real numbers satisfying $x + y + z = 1$. Find the exact minimum value of $\frac{1}{x} + \frac{9}{y} + \frac{16}{z}$.

a. 64

b. 26

c. 72

d. 61

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12 Let x, y, z be positive real numbers satisfying $x + y + z = 9$. Find the exact minimum value of $\frac{4}{x} + \frac{9}{y} + \frac{16}{z}$.

a. 29

b. 18

c. 6

d. 9

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